

$$\begin{aligned}
 x^2 + y^2 &= 1 \\
 1 - x^2 - y^2 &= 0 \\
 \Rightarrow \frac{1 - x^2 - y^2 + 2yi}{(1 - x)^2 + y^2} \\
 &= \frac{2yi}{(1 - x)^2 + y^2}
 \end{aligned}$$

which is purely imaginary.

QUESTION THREE

- a. If $u = q^2 \operatorname{cis} \frac{3\pi}{4}$ and $v = q^3 \operatorname{cis} \frac{\pi}{3}$, write $\frac{u}{v}$ in the form $r \operatorname{cis} \theta$.

$$\begin{aligned}
 \frac{u}{v} &= \frac{q^2 \operatorname{cis} \left(\frac{3\pi}{4} \right)}{q^3 \operatorname{cis} \left(\frac{\pi}{3} \right)} \\
 &= \frac{1}{q} \operatorname{cis} \left(\frac{5\pi}{12} \right)
 \end{aligned}$$